

$$\begin{array}{ccc}
 [X, Y] & \xrightarrow{\quad \quad \quad \equiv \quad \quad \quad} & [X, Y] \\
 \downarrow \eta_{[X, Y]} & & \downarrow [X, \eta_Y] \\
 & (ii) & [X, U^T F^T Y] \\
 & & \uparrow [\eta_X, U^T F^T Y] \\
 U^T F^T [X, Y] & \xrightarrow{U^T \psi_{X, Y}} U^T [F^T X, F^T Y]_T \xrightarrow{\quad \quad \quad \equiv \quad \quad \quad} & [U^T F^T X, U^T F^T Y]
 \end{array}$$